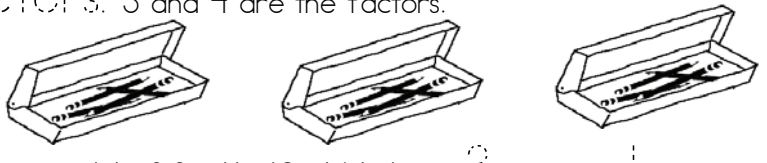
Multiplication is a faster way to add. When we multiply, we take a shortcut. Here's an example: Jamie has 3 boxes of pencils. Each box contains 4 pencils. Jamie has 12 pencils total.

Instead of saying 4 + 4 + 4 = 12, we can say 3 x 4 = 12. She has 3 groups of 4. The two numbers we multiply together are called factors. 3 and 4 are the factors.

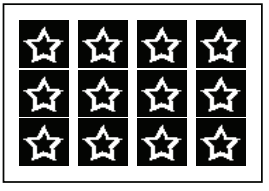


This is a model of 3 x 4 =12. We have 3 equal groups of four with a product of 12. A product is the answer to a multiplication problem.

Another model that we can use to show 3 x 4= 12 is an array. This is an example of an array showing 3 x 4 = 12. We have three rows with 4 in each row.

4 is a factor.

3 is a factor.

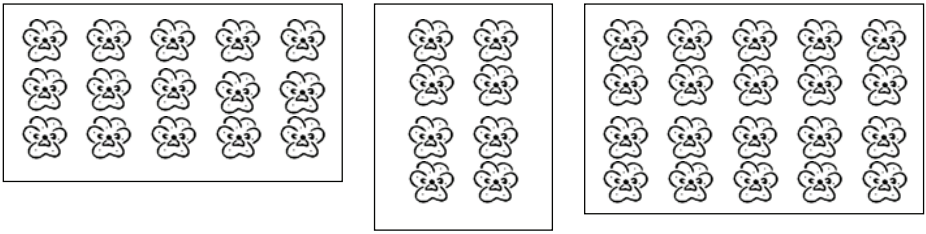


The product is 12.

Helping Hand Hint: If you see the words "each" or "groups of" in a problem, the operation may be multiplication.

Arrays:

3x5=15 4x2=8 4x5=20



3 and 5 are factors, 4 and 2 are factors,

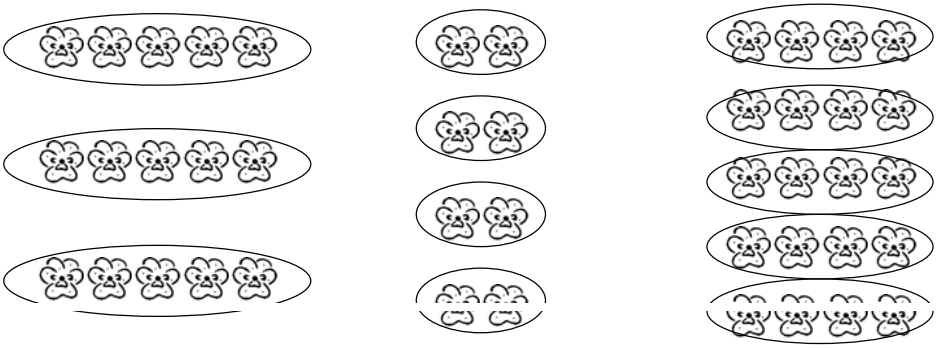
5 and 4 are factors

15 is the product, 8 is the product,

20 is the product

These are models of the same multiplication facts. Look at the number sentences.

3x5=15 4x2=8 4x5=20



Factor

3x4

Product

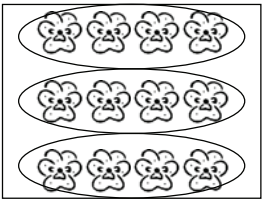
3x5

Array of 4x2

Multiplication

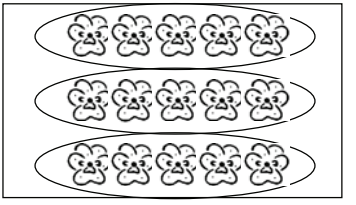
5x4

Answer to a multiplication problem

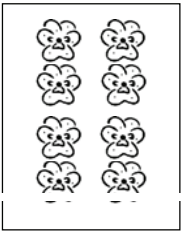


The numbers you multiply to get the product

An operation that is a faster way to add



20



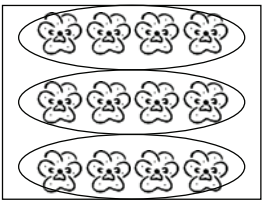
Draw a line to the correct answer.

Factor

3x4

Product

Answer to a multiplication problem



The numbers you multiply to get the product

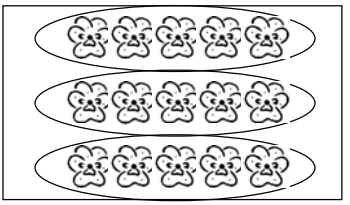
3x5

Array of 4x2

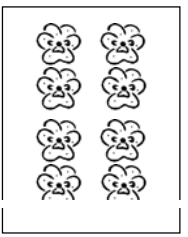
Multiplication

5x4

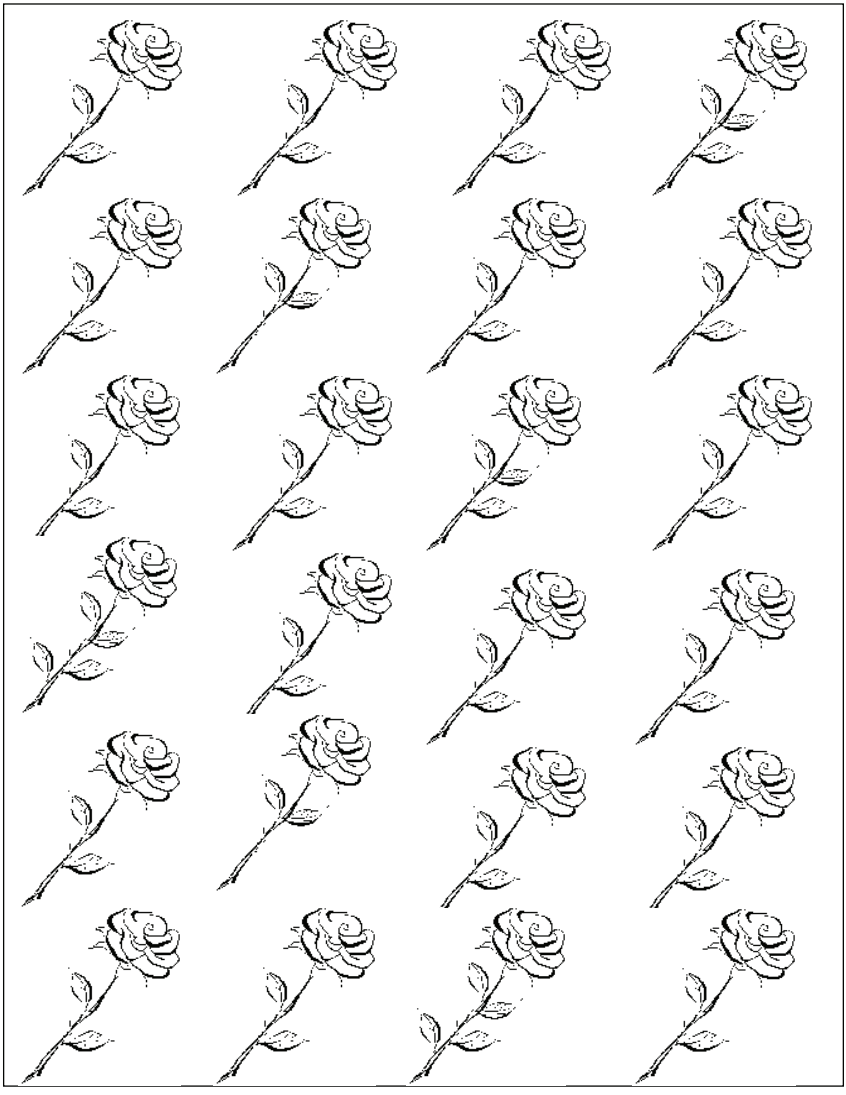
An operation that is a faster way to add



20



Here is an array of flowers. Each row has equal amounts in them. One flower in every row is different. Circle the flower in each row that is different. Then in the space below write the number of flowers there are in total. _____



When we multiply we find the total number in each group. You can use an array or number sentence to find the product of a problem.

Josh has three bags of golf balls. There are five balls in each bag. How many golf balls does Josh have in all?

To solve this problem, we need to decide how many groups there are and how many are in each group.

There are three bags or groups.

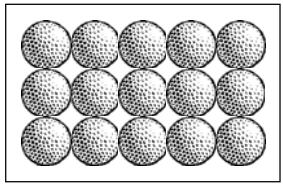
There are five balls in each group.

Josh has 3 groups of 5 and can show this three ways:

Number sentence

3x5=15

Array

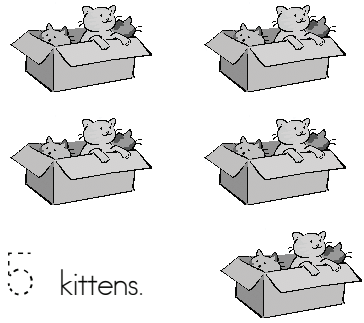


groups

The product is fifteen. Josh has 3 groups of 5. He has fifteen golf balls in all. The number sentence is 3x5=15

Read and trace the dotted keywords.

John has five boxes. Each box has three kittens in it. How many kittens does he have?



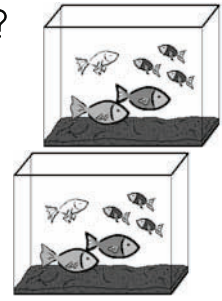
How many boxes are there? Five

How many kittens are in each box? Three

The number sentence would be 5x3=15

How many kittens does John have? John has 15 kittens.

Paige has two aquariums. She wants to put six fish in each aquarium. How many fish does Paige need to buy?



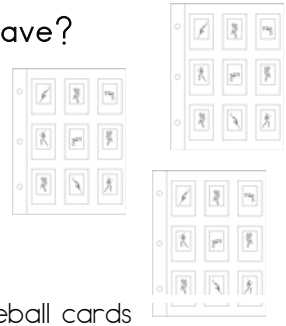
How many aquariums are there? Two

How many fish does she want in each aquarium? Six

The number sentence would be 2x6=12

How many fish does Paige need to buy? Paige would buy 12 fish.

Dallas has three pages. Each page has nine baseball cards on it. How many baseball cards does Dallas have?



How many pages are there? _____

How many cards are on each page? _____

The number sentence is _____

How many cards does Dallas have? Dallas has 27 baseball cards

Greg has a bookcase with three shelves. He is going to put five books on each shelf. How many books would he have?

_____X_____=_____books

Jessica has two pencil boxes. She has four pencils in each box. How many pencils does Jessica have in all?

_____X_____=_____pencils

Lane has four bags for his marbles. He has 7 marbles in each bag. How many marbles does Lane have all together?

_____X_____=_____marbles

Shari has five boxes. She puts two kittens in each of the boxes to give away. How many kittens will Shari give away?

_____X_____=_____kittens

Greg has a bookshelf with three shelves in it. He's going to put five books on each shelf. How many books would he have?

_____X_____=_____books
3 5 15

Jessica has two pencil boxes. She has four pencils in each box. How many pencils does Jessica have in all?

_____X_____=_____pencils
2 4 8

Lane has four bags for his marbles. He has 7 marbles in each bag. How many marbles does Lane have all together?

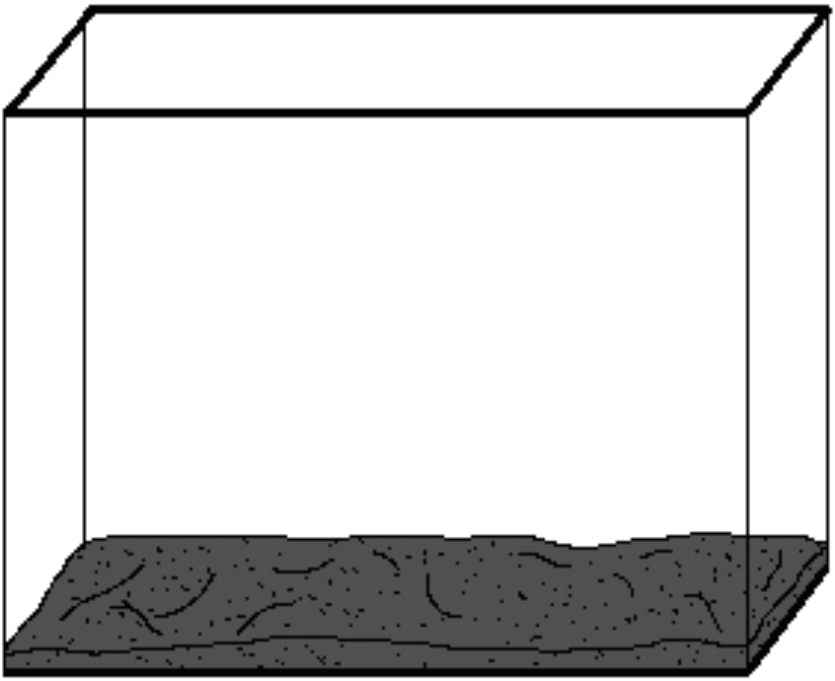
_____X_____=_____marbles
4 7 28

Shari has five boxes. She puts two kittens in each of the boxes to give away. How many kittens will Shari give away?

_____X_____=_____kittens
5 2 10

Here's one of Paige's four aquariums. She put six fish in each.

Draw six different kinds of fish that Paige could put in her aquarium. Draw anything else you think would look good in your aquarium. Then write the number sentence for the total number of fish in Paige's four aquariums.



_____ x _____ = 24 fish

Anytime we multiply a number by zero, the product will always be zero. Example: 4x0=0. That would be like saying that there were four bags with nothing in them. So any multiplication number sentence that has zero as one of the factors, the product will always be zero.

When you multiply a number by one, the answer will always be the same as the other factor. Example: 5x1=5. This would be the same as five boxes with one puppy in each box. So, whenever one of the factors is one, the answer will be the same as the other factor.

Practice these multiplication facts:

0x0=0 0x1=0 0x2=0 0x3=0

0x4=0 0x5=0

1x0=0 1x1=1 1x2=2 1x3=3

1x4=4 1x5=5

Read and trace the dotted keywords.

Anytime you multiply a factor by zero, the product will be zero. 4x0=0

When you multiply a factor by one, the product will always be the same number as the other factor. 5x1=5

0x0=0

0x1=0

0x2=0

0x3=0

0x4=0

0x5=0

1x0=0

1x1=1

1x2=2

1x3=3

1x4=4

1x5=5

Read and trace the dotted keywords and the number sentences.

0x1=_____

1x12=_____

1x6=_____

0x11=_____

1x2=_____

1x5=_____

1x10=_____

0x2=_____

0x4=_____

1x11=_____

0x8=_____

0x12=_____

1x4=_____

1x3=_____

0x3=_____

1x1=_____

0x5=_____

0x7=_____

1x7=_____

0x6=_____

Write the correct answers in the blanks.

0x1=_____
0

1x12=_____
12

1x6=_____
6

0x11=_____
0

1x2=_____
2

1x5=_____
5

1x10=_____
10

0x2=_____
0

0x4=_____
0

1x11=_____
11

0x8=_____
0

0x12=_____
0

1x4=_____
4

1x3=_____
3

0x3=_____
0

1x1=_____
1

0x5=_____
0

0x7=_____
0

1x7=_____
7

0x6=_____
0

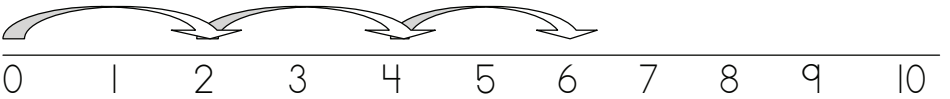
How many new words can you make out of the words:

Multiplication Facts

multiplication

Make new words using the words Multiplication Facts.

When we multiply a number by two we are also skip counting by twos. Look at this number line:

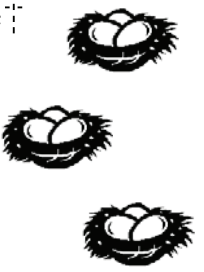


We are also putting objects in groups of twos. We have two groups of three.

The number sentence for these two models is 2x3=6.

The same is true for multiplying by three. It is also skip counting by threes. Look at the number line.

This time we are putting objects in groups of threes. We have three groups of three.



The number sentence is 3x3= 9.

0 1 2 3 4 5 6 7 8 9 10

2x0=0

2x1=2

2x2=4

2x3=6

2x4=8

2x5=10

0 1 2 3 4 5 6 7 8 9 10

3x0=0

3x1=3

3x2=6

3x3=9

3x4=12

3x5=15

sentences using the number line.

0 1 2 3 4 5 6 7 8 9 10

3x1=_____

2x1=_____

2x5=_____

3x0=_____

2x0=_____

2x4=_____

3x2=_____

3x5=_____

2x2=_____

3x3=_____

3x4=_____

2x3=_____

line to help you find the answer.

0 1 2 3 4 5 6 7 8 9 10

3x1=_____
3

2x1=_____
2

2x5=_____
10

3x0=_____
0

2x0=_____
0

2x4=_____
8

3x2=_____
6

3x5=_____
15

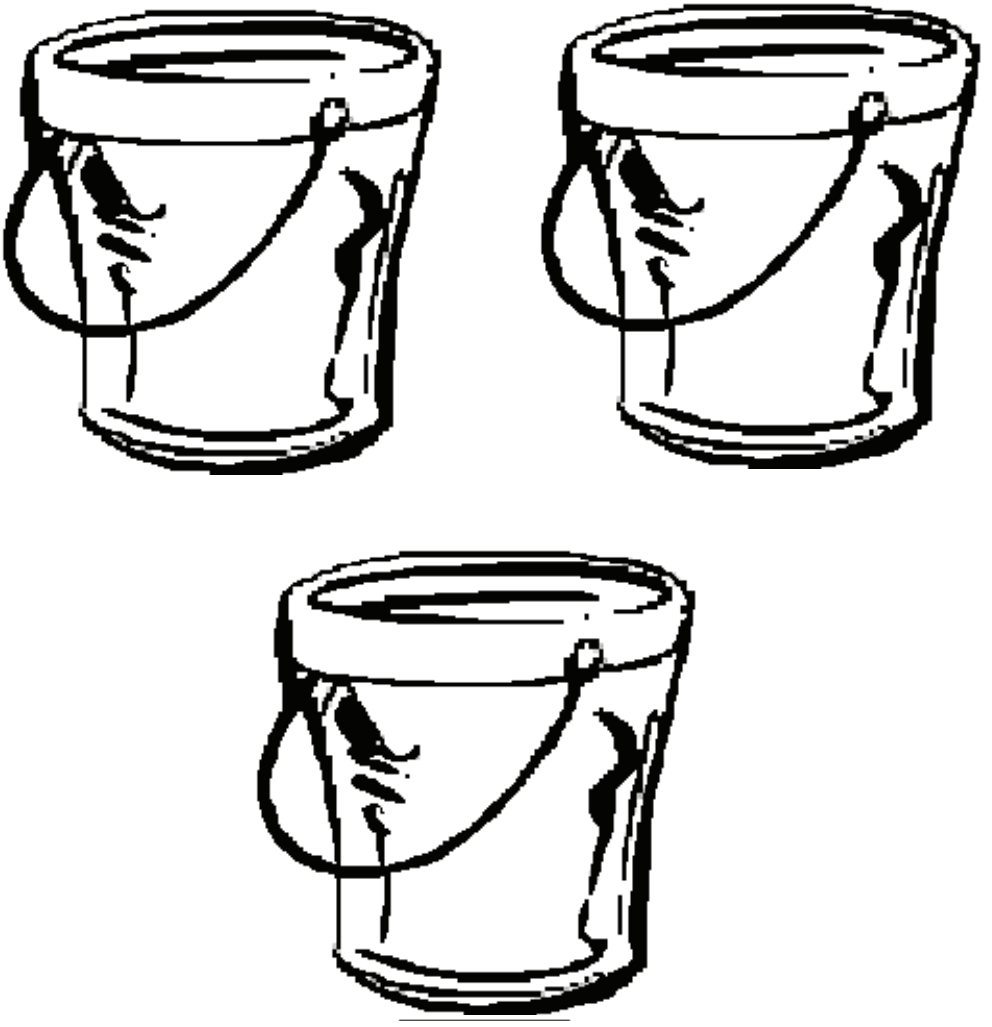
2x2=_____
4

3x3=_____
9

3x4=_____
12

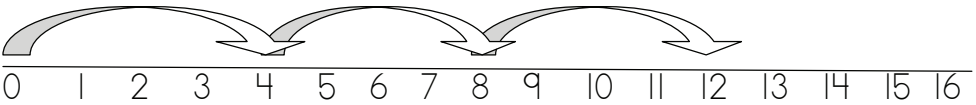
2x3=_____
6

Here are four buckets. Can you fill each bucket with an equal amount? Draw what you would like in each bucket.



equal amount.

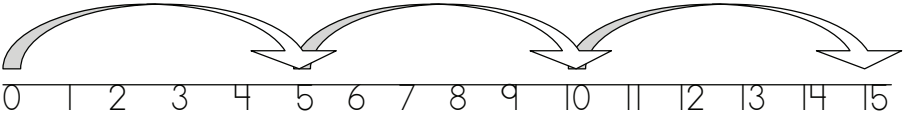
When we multiply a number by four we are skip counting by fours. Look at the number line:



We are putting objects in groups of four: Four groups of three.

The number sentence for this 4x3=12

Multiplying by five is just like skip counting by fives. Look at the number line.



This is five groups of three or 5x3=15

Trace the dotted number sentences and practice using the number line.

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

4x0=0

4x1=4

4x2=8

4x3=12

4x4=16

4x5=20

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

5x0=0

5x1=5

5x2=10

5x3=15

5x4=20

5x5=25

Trace the dotted number sentences and practice using the number line.

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

4x5=_____

5x5=_____

5x1=_____

4x3=_____

5x4=_____

5x0=_____

4x0=_____

4x1=_____

5x2=_____

5x3=_____

4x4=

4x2=

Fill in the blank with the product.
Use the number line for help.

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

4x5=_____
20

5x5=_____
25

5x1=_____
5

4x3=_____
12

5x4=_____
20

5x0=_____
0

4x0=_____
0

4x1=_____
4

5x2=_____
10

5x3=_____
15

4x4=
16

4x2=
8

Multiplying by five is the same as counting by fives. You have five fingers on each hand. Trace around your hand. What kind of different animals can you make when you trace your hand with five fingers?

Multiplication is a faster way to add.

When we multiply, we take a shortcut. The two numbers that we multiply are called factors and the answer is the product.

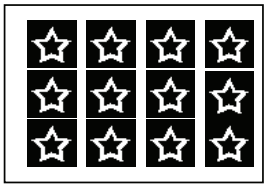
We put objects into equal groups when we multiply.

<u>Hint:</u> If you see the words "each" or "groups of" in a problem, the operation may be multiplication.

An array is a model that we can use to show multiplication. This is an example of an array showing the number sentence 3 x 4 = 12

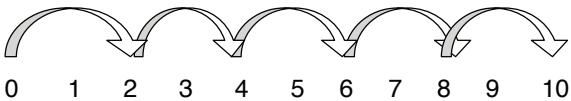
4 is a factor.

3 is a factor.



The product is 12.

Number lines help us when we multiply. This number line is skip counting by twos.



0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

0x0=0 1x0=0 2x0=0 3x0=0 4x0=0 5x0=0

0x1=0 1x1=1 2x1=2 3x1=3 4x1=4 5x1=5

0x2=0 1x2=2 2x2=4 3x2=6 4x2=8 5x2=10

0x3=0 1x3=3 2x3=6 3x3=9 4x3=12 5x3=15

0x4=0 1x4=4 2x4=8 3x4=12 4x4=16 5x4=20

0x5=0 1x5=5 2x5=10 3x5=15 4x5=20 5x5=25

Practice these multiplication facts and trace the answers.

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

5x4=___ 1x0=___ 3x0=___ 0x5=___ 0x0=___

1x2=___ 2x5=___ 0x1=___ 1x4=___ 3x3=___

4x3=___ 5x5=___ 4x2=___ 2x0=___ 2x3=___

0x3=___ 4x4=___ 1x3=___ 0x4=___ 3x2=___

3x4=___ 4x5=___ 5x0=___ 4x1=___ 5x2=___

5x1=___ 3x5=___ 2x4=___ 1x5=___ 1x1=___

Give the correct answer for each number sentence.

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

5x4=___ 20	1x0=___ 0	3x0=___ 0	0x5=___ 0	0x0=___ 0
1x2=___ 2	2x5=___ 10	0x1=___ 0	1x4=___ 4	3x3=___ 9
4x3=___ 12	5x5=___ 25	4x2=___ 8	2x0=___ 0	2x3=___ 6
0x3=___ 0	4x4=___ 16	1x3=___ 3	0x4=___ 0	3x2=___ 6
3x4=___ 12	4x5=___ 20	5x0=___ 0	4x1=___ 4	5x2=___ 10
5x1=___ 5	3x5=___ 15	2x4=___ 8	1x5=___ 5	1x1=___ 1

2 5 9 3 4 7 9 10 4 12 6

4 1 7 5 3 3 4 12 6

12 5 2 11 13 14 1 4 12

13 8 5 15 9 6 14 16 5 7 1

---

1 2 3 4 5 6 7 8 9 10 11 12 13
14 15 16

s m t i u g p q l y b n e
r d c

Decode the sentence using the key at the bottom.

Jamie made five cherry pies. Each pie has two cups of cherries in it How many of cups of cherries did she use?

Draw the number of cups in each pie. Write the number sentence.

_____x_____=_____

Jamie used ____ cups of cherries to make her pies.

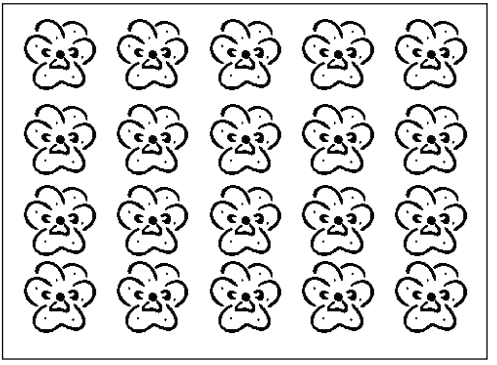
Which number sentence tells me how many flowers there are in this array?

3x5=15

4x3=12

5x5=25

4x5=20



for both problems.

Jamie made five cherry pies. Each pie has two cups of cherries in it How many of cups of cherries did she use?

Draw the number of cups in each pie. Write the number sentence.

_____x_____=_____
2 5 10

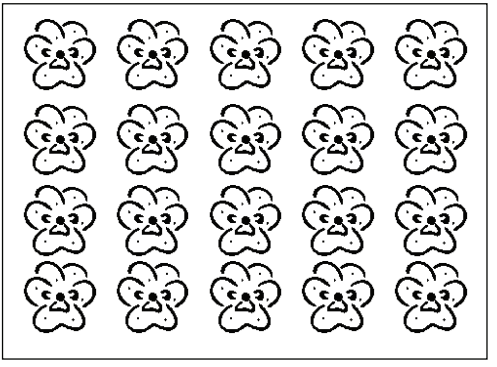
Jamie used _____ cups of cherries to make her pies.
10

Which number sentence tells me how many flowers there are in this array?

3x5=15

4x3=12

5x5=25



4x5=20

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